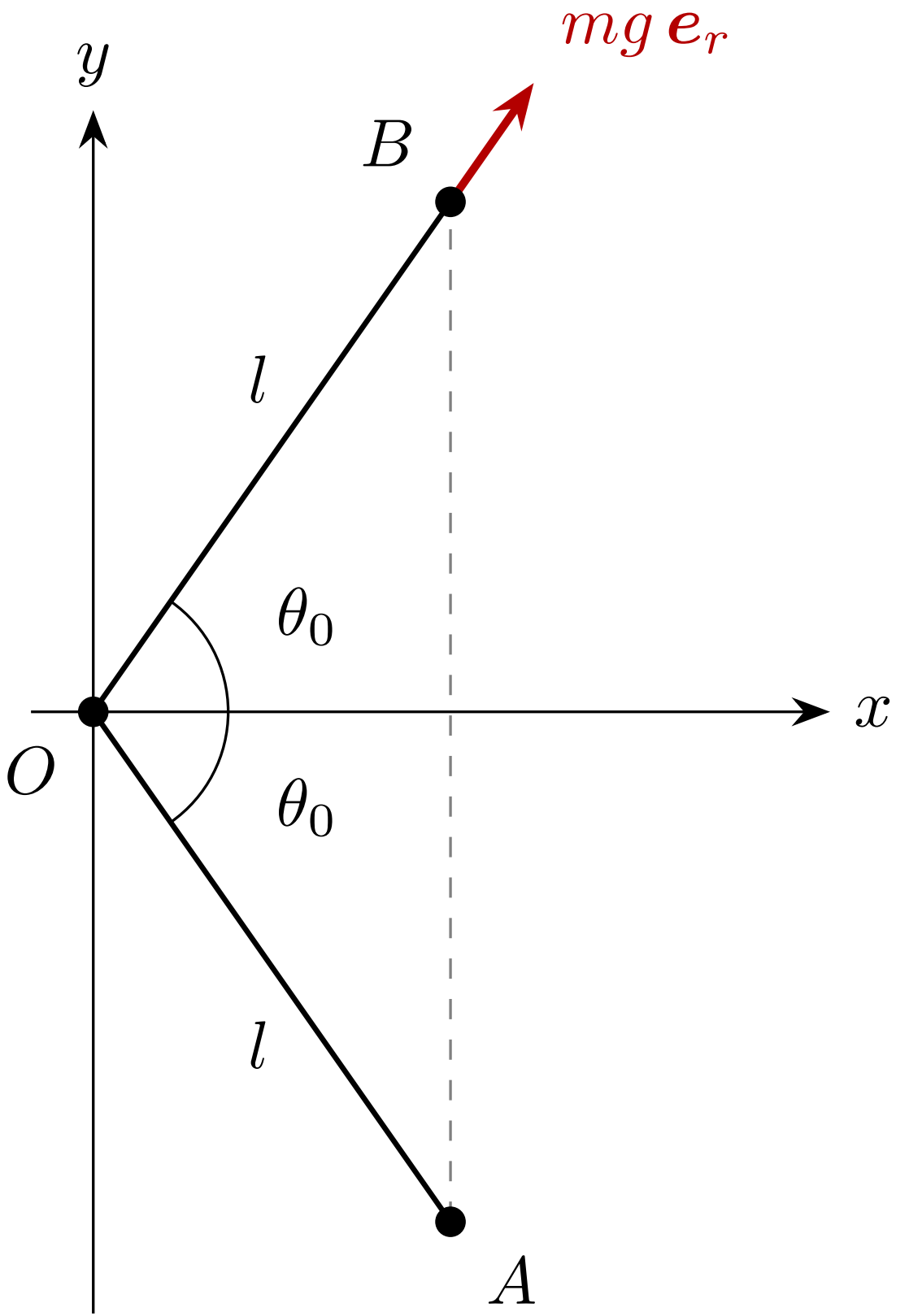
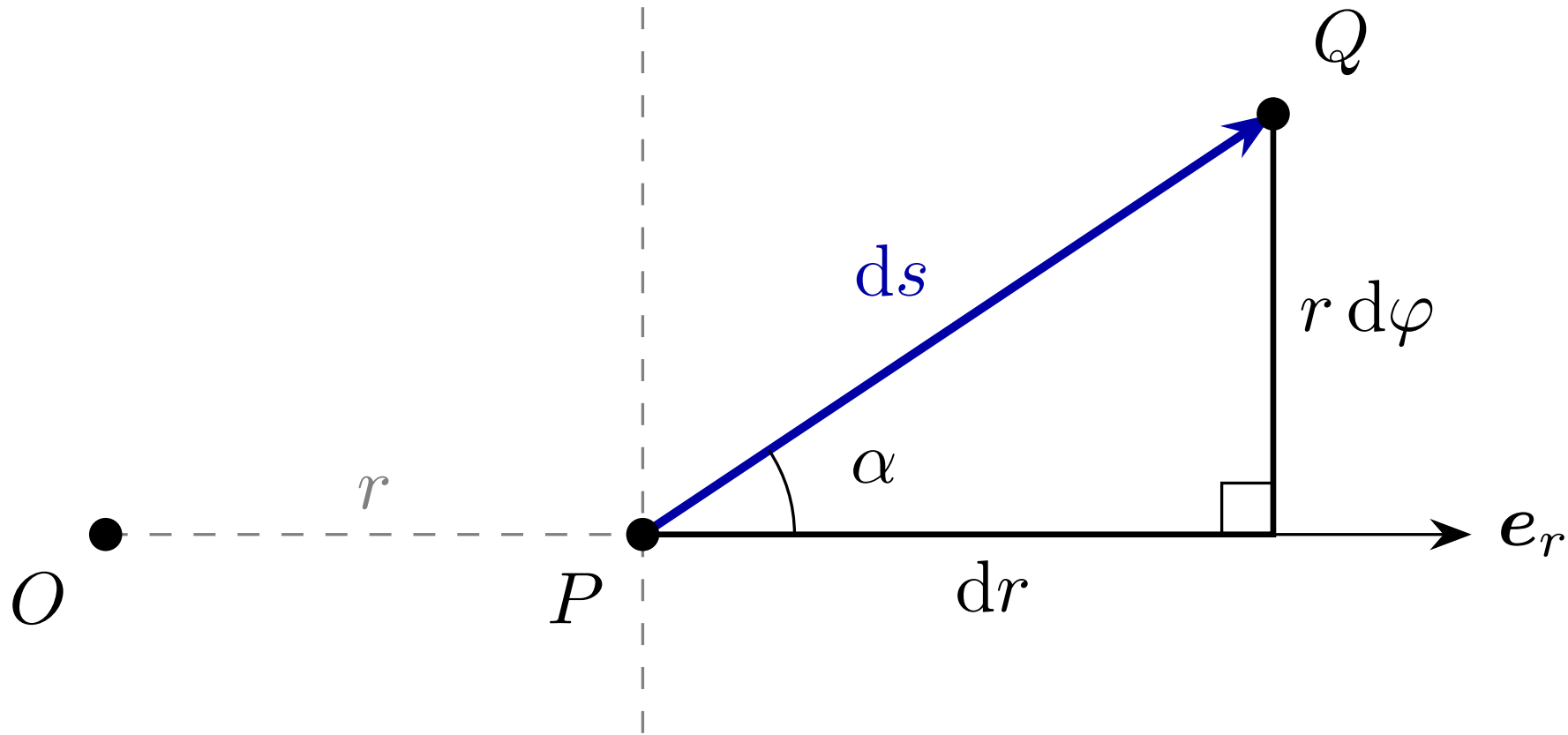
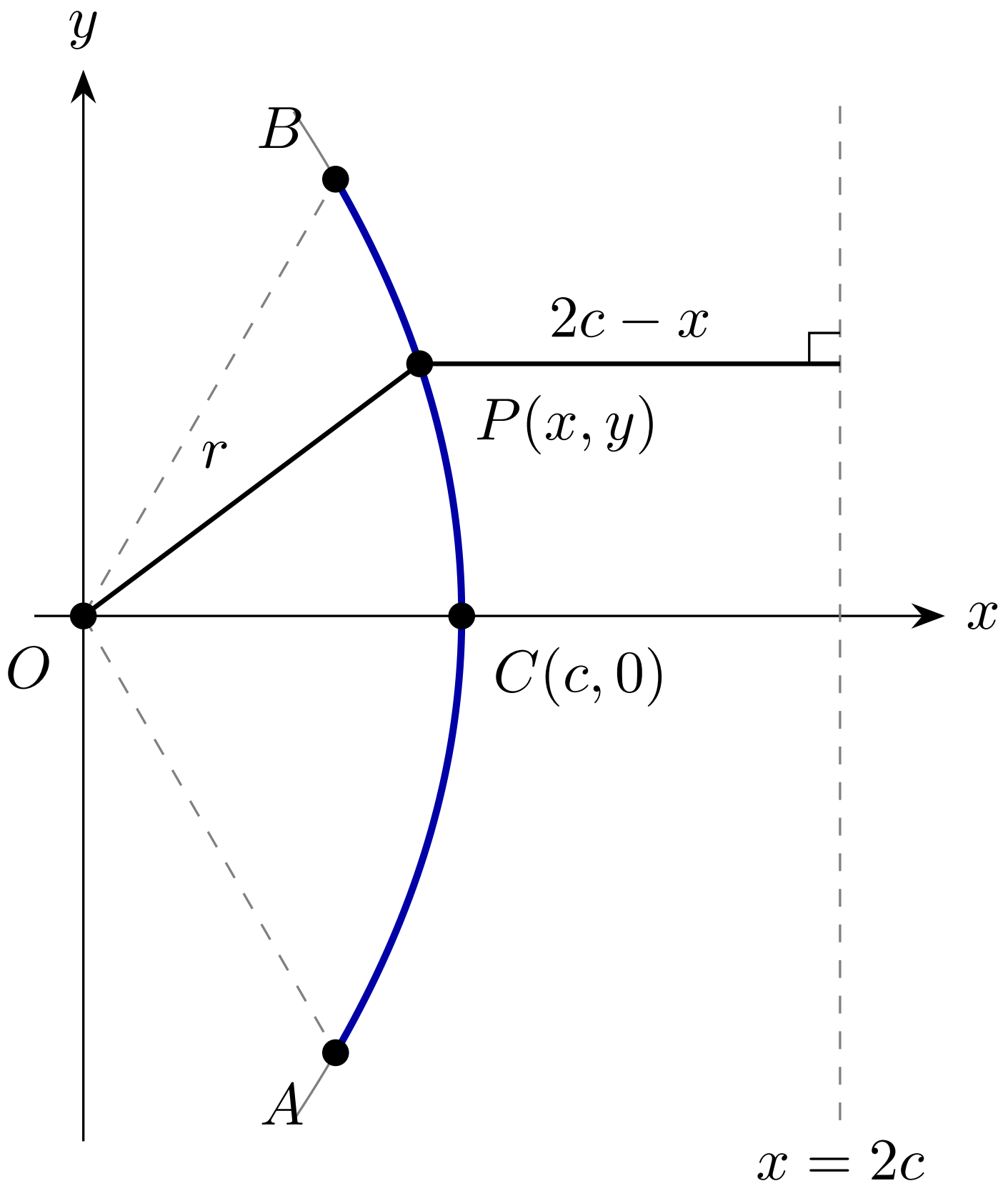
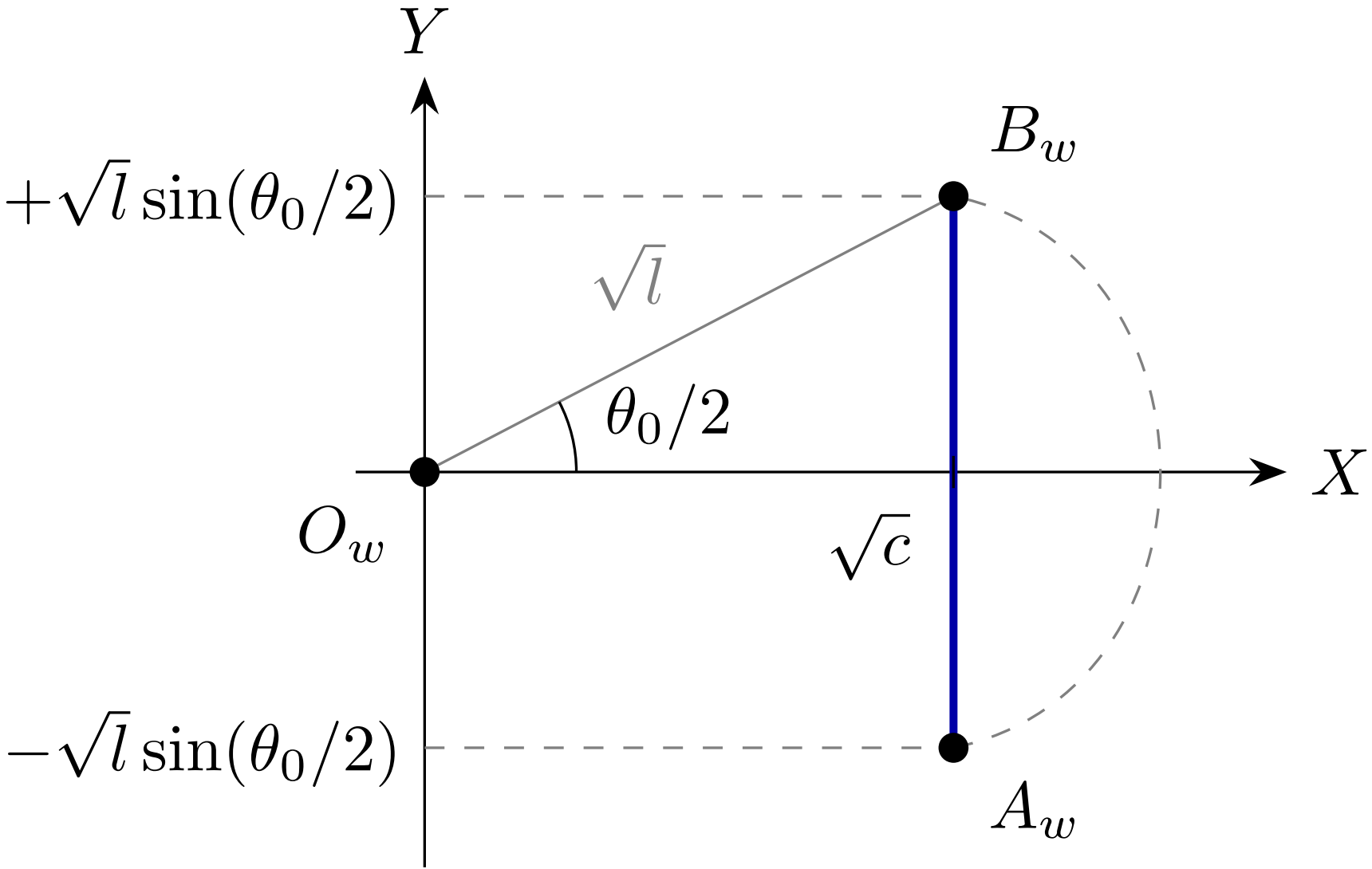
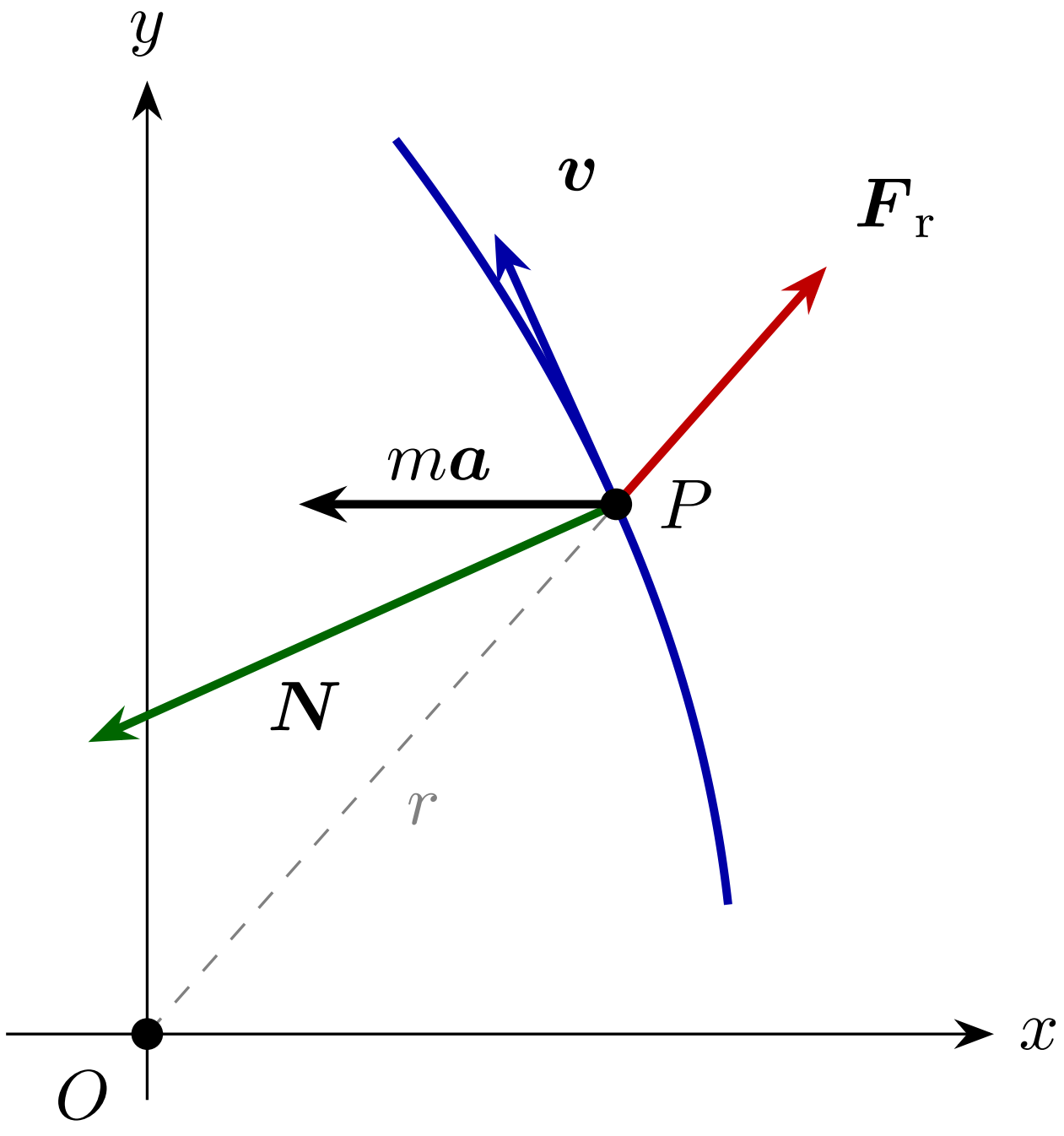

$mg\,\boldsymbol{e}_r$
$y$
$B$
$l$
$\theta_0$
$x$
$O$
$\theta_0$
$l$
$A$


Q
ds
r dφ
α
r
O
P
dr
e_r


y
B
2c − x
P(x, y)
r
x
O
C(c, 0)
A
x = 2c


$Y$
$B_w$
$+\sqrt{l}\sin(\theta_0/2)$
$\sqrt{l}$
$\theta_0/2$
$X$
$O_w$
$\sqrt{c}$
$-\sqrt{l}\sin(\theta_0/2)$
$A_w$


y
v
F_r
ma
P
N
r
O
x